

Cloud Computing Overview

Cloud Computing is the delivery of computing services such as servers, storage, databases, networking, software, and analytics over the internet. Instead of purchasing and maintaining physical hardware, organizations can access resources on demand from cloud providers.

Cloud computing offers several benefits including scalability, flexibility, cost efficiency, and high availability. Businesses can quickly scale resources up or down based on demand without investing heavily in infrastructure.

There are three primary cloud service models. Infrastructure as a Service (IaaS) provides virtualized computing resources such as servers and storage. Platform as a Service (PaaS) offers a development environment where developers can build and deploy applications without managing underlying infrastructure. Software as a Service (SaaS) delivers software applications directly to users through web browsers.

Cloud deployment models include Public Cloud, Private Cloud, and Hybrid Cloud. Public Cloud services are offered by third-party providers and shared among multiple organizations. Private Cloud environments are dedicated to a single organization and provide greater control and security. Hybrid Cloud combines both approaches to achieve flexibility and compliance.

Major cloud providers include Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP). These providers offer a wide range of services including virtual machines, storage systems, databases, artificial intelligence tools, and security solutions.

Cloud security is a critical concern. Organizations must implement identity management, encryption, network security, monitoring, and compliance controls to protect data and applications. Shared responsibility models define security responsibilities between cloud providers and customers.

Cloud computing supports digital transformation by enabling faster innovation, global scalability, and improved operational efficiency. Organizations use cloud platforms for web hosting, application development, disaster recovery, big data analytics, and machine learning workloads.

As businesses continue modernizing their IT infrastructure, cloud computing remains one of the most important technologies driving growth, agility, and competitiveness in the digital economy.